Task F3 — Deployment Recommendation

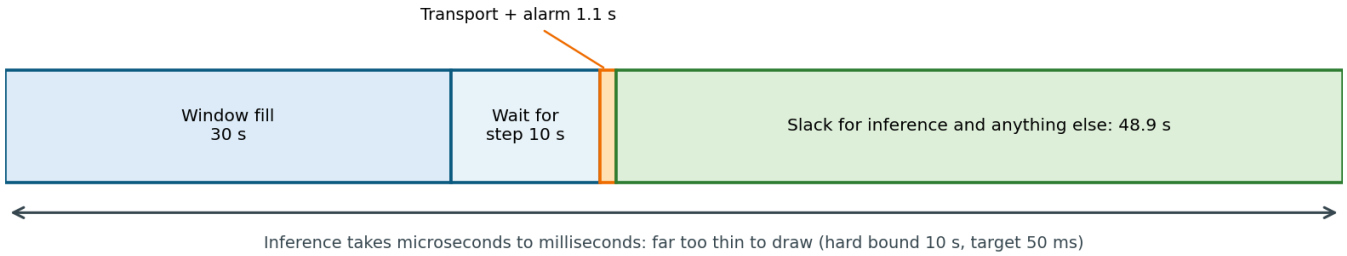
Which model for the 85 trucks? | LogiEdge | FreightBridge Logistics Pvt. Ltd. | Component F, Module 6

At a glance

Question	Answer
Which model?	**M1, the FP32 baseline**, retrained with vibration-only runs added (Section 5), on all 85 Raspberry Pi 5 trucks
Why not INT8 or pruned?	On this device and model size they save nothing that matters and cost robustness: M2 and M3 raise 1.3 to 1.6 times the false alarms of M1 under noise (12σ), and add calibration and pruning steps to every update
Latency	Budget 10 s per inference (hard). The operation count suggests microseconds (to be measured), so the 90 s SLA is consumed by the 30 s window and 10 s step, not by inference
Flash and SRAM	No constraint on the Pi 5. Even on the STM32H743 fallback, M1 uses 0.16% of 2 MB flash and 0.02% of 1 MB RAM
Critical recall	All variants exceed 95%: 100% on the validation split (17/17 Critical) and on 1,195 independent Critical windows (95% lower bound 99.7%)
Must fix first	**Every variant calls 100% of vibration-only faults Normal** (bearing wear with normal temperature); the training data had no such case. Adding it fixes this (Section 5).

1 The 90 s SLA as a per-inference latency budget

90 s SLA: from a fault signature appearing in the sensor data to the alert



Item	Time	Source
SLA: fault signature in the data to the alert	90 s	Task A1
Window fill (evidence can take the whole window)	up to 30 s	Task C2
Wait for the next window step	up to 10 s	Task C2
Preprocessing, MQTT and local alarm	about 1.1 s	My allowance; alarm 0.1 s from Task A1
Left for inference	**about 48.9 s**	90 − 41.1
Hard per-inference bound	**10 s**	One inference per 10 s step; slower than that and windows queue up
Engineering target	50 ms	Task A1 allowance

- The binding limit is the 10 s step, not the 48.9 s of slack. Each variant is about 750 multiply-accumulates (366 for M3), so I expect the work to take microseconds (an estimate from the operation count, not yet measured). Latency should not separate the variants.
- Measured in simulation, the first alert came about 20 s after fault onset for all three variants (30 s once for M2), well inside 90 s.
- The TFLite latencies on your laptop and a Pi 5 are still to be measured (Task F2). The decision rule: any variant whose p95 is under 50 ms qualifies.

2 Flash and SRAM for the chosen device

	M1 FP32	M2 INT8	M3 pruned INT8
Stored numbers (the model data in flash)	3,212 B	1,568 B	910 B
Estimated .tflite file	4.4–5.4 KB	3.2–4.4 KB	2.5–3.7 KB
Peak activation memory (SRAM or RAM)	192 B	48 B	31 B

Device	Flash or storage	SRAM or RAM	M1 uses
Raspberry Pi 5, 8 GB + AI HAT+ (chosen in Task B1)	microSD or NVMe, tens of GB (assumed 32 GB)	8 GB LPDDR4X	About 0.00001% of storage and under 0.0001% of RAM
STM32H743 MCU (Task B1 Option 3, fallback)	Up to 2 MB (1 MB on the xG parts)	Up to 1 MB: 192 KB TCM + 864 KB user SRAM	3.2 KB = 0.16% of 2 MB flash; 192 B = 0.02% of RAM

The STM32H743 figures are from ST's datasheet page; the Pi 5 storage size is my assumption. TFLite and interpreter code and your application add to these numbers, and I did not measure them.

If the production model is the 280 KB INT8 model of Task E3: on the Pi 5 it is still trivial, but on the 1 MB-flash MCU 280 KB would be 28% of flash, and an FP32 version (about four times larger, roughly 1.1 MB) would not fit. INT8 would then be needed. That is the one case that flips the decision.

3 Class 2 (Critical) recall

Test	M1	M2	M3	Needs
Validation split (17 Critical windows)	100.0%	100.0%	100.0%	> 95%
Independent runs (1,195 Critical windows)	100.0%	100.0%	100.0%	> 95%
With noise of 8σ added	98.8%	98.9%	98.7%	> 95%
With noise of 12σ added	95.4%	96.1%	95.1%	> 95%

- **M1 meets the target.** 100% on both test sets. The 95% lower bound is 81.6% on the 17-window split (too few windows to prove more) and 99.7% on the independent runs, which is the stronger evidence.
- **The required D1 model would not have passed on the split:** 16 of 17 Critical windows, 94.1%. That is why M1 is the onset-trained model.
- **Recall is by construction Critical = the combined fault.** Vibration-only faults are outside that definition and are missed (Section 5).

4 Decision

Criterion	M1 FP32	M2 PTQ INT8	M3 pruned + INT8
90 s SLA and 10 s bound	Met, by orders of magnitude	Met	Met
Flash and SRAM	Fits	Fits	Fits
Critical recall above 95%	Yes, 100%	Yes, 100%	Yes, 100%
False alarms under noise (12σ)	4.2% (lowest)	6.6%	5.4%
Extra steps on every update	None	Calibrate on current data	Prune, fine-tune, calibrate, validate
Update download (about)	4.4–5.4 KB	3.2–4.4 KB	2.5–3.7 KB
Energy per inference	To be measured	To be measured	To be measured
Verdict	**Deploy**	Only if memory or bandwidth binds	Not worth the extra work

- **Savings are negligible:** M3 downloads about 2 KB less than M1, which across 85 trucks is 0.17 MB or about ₹0.02 per update at ₹0.10/MB.
- **Energy is negligible too, even pessimistically.** If each inference ran the whole 7.5 W Pi for 1 ms (far above what an operation count suggests), a day's 8,640 inferences would use 8,640 × 7.5 mJ = 65 J = 0.018 Wh, which is 0.01% of the 180 Wh the device uses in 24 hours.
- **INT8 buys no accuracy and costs robustness here.** Quantising Normal-class inputs onto one or two levels is the price of the large input range; with no memory or speed problem to solve, it is a risk with no benefit.
- **What would change the decision:** a microcontroller target with a large model (above), measured Pi 5 latency or energy that proves much worse than expected for FP32, or a bandwidth cost that mattered (at ₹0.10/MB it does not).

5 Fix before the 85-truck rollout: vibration-only faults

Test windows (--anomaly vibration)	Flagged Warning or Critical	False alarms on pre-onset windows
D1 models, all of M1, M2, M3 and the required model	0% (87 of 87 predicted Normal)	0%
Retrained with vibration-only runs added (labelled Warning, Task C scheme B)	100.0%	0.0%

- A compressor bearing-wear signal 75σ above normal, with normal temperature, is invisible to the model, because the Task D1 data only had Normal, temperature drift and combined runs. By the Task C table that is an “early refrigeration anomaly” or a unit-failure signature, so it must not be missed.
- Adding the vibration-only runs removes the gap, with no change on the other tests (validation 100.0%, independent runs 100.0%, Critical recall 100.0%). `vibration_gap_check.py` reproduces this.
- Recommended path: add vibration-only runs to the training set, retrain M1, rerun the tests and the E1 reference, then roll out via the shadow-mode process of Task E3.

6 Items to confirm

- **Class for vibration-only faults.** I labelled them Warning (Task C scheme B). If you consider them Critical, tell me.
- **Measured latency and energy.** Run Task F2's benchmark and send benchmark_results.json; the decision rule in Section 1 then closes.
- **Device.** I assumed the Raspberry Pi 5 + AI HAT+ from Task B1. If trucks use the MCU option, the memory section matters and the 280 KB model becomes the question.
- **Production model.** If your real model is the 280 KB INT8 one, FP32 is about four times larger; tell me its size and I will redo the memory and bandwidth numbers.